

ASSIGNMENT 1

Subject: Software Testing

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1. Difference between Static Testing and Dynamic Testing. Give 2 examples of techniques used in each type.

Static Testing

- Process of finding defects without executing the application code.
- Involves reviewing documents, designs and code to ensure correctness.
- Prevent defects in the early stages of SDLC.
- It's a type of preventive approach.
- It's a part of verification.
- Saves time and cost.

Techniques:

- **Requirement Review:** check if requirements are complete, missing or incorrect.
- **Design Review:** verify design matches requirements and it's feasible.

Dynamic Testing

- Process of finding defects by executing the application code.
- Involves running the system and checking outputs against expected results.
- Detects defects during execution. So it's a type of corrective approach.
- It's a part of validation.
- Finds logical or functional errors.
- Checks how software behaves in real time.

Techniques:

- **Black Box Testing**
 - Tests what the system does.
 - Focuses on input and output.
 - Testing based on requirements without knowing internal code.
 - Testers do not need programming knowledge.
 - Equivalence partitioning, boundary value analysis, decision table testing, state transition testing, error guessing are the types of techniques in black box testing.

- White Box Testing
 - Focuses on logic, paths, conditions.
 - Testing with knowledge of internal code.
 - Requires programming knowledge, usually done by developers.
 - Statement coverage, branch coverage, path coverage, condition coverage, mutation testing are the types of techniques in white box testing.

2. Explain the principles of software testing. Why is early testing considered cost effective in software development?

There are 7 principles of software testing:

1. **Testing shows presence of defects:** It cannot prove that software is 100% bug free.
2. **Exhaustive testing is impossible:** you can't test everything.
3. **Early testing saves time and cost:** start testing from the requirement phase.
4. **Defect clustering:** most defects are found in a few modules.
5. **Pesticide paradox:** running the same test cases again and again finds fewer new bugs.
6. **Testing is context dependent:** testing approach depends on type of application.
7. **Absence of errors is a fallacy:** even if no bugs are found, software is useless if it doesn't meet user needs.

Early testing is cost effective because:

- Defects found in early stages are cheaper to fix.
- Saves time and rework cost.
- Improves product quality.

3. Describe the Waterfall Model and the V-Model in SDLC. How do they differ in handling Verification and Validation?

Waterfall Model

- Linear, step by step model.
- Each phase must finish before the next phase starts.
- Phases: Requirements → Design → Implementation → Testing → Deployment → Maintenance .
- Testing happens after development is completed.
- Less flexible to changes.
- Best for small projects with clear requirements.

V-Model

- Extension of waterfall model.
- Shaped like a “V”, development phases at left side and corresponding testing phases at right side.
- Testing activities are planned parallel to each development phase.
- So defects are found early.
- Suitable for projects requiring high quality.

Difference in handling Verification and Validation:

Waterfall Model

- Verification mainly through reviews in early phases.
- Validation done after development is complete.
- Defects found late.

V-Model

- Verification done at every development stage.
- Validation planned parallel to each phase.
- Defects found early.

4. What are the key roles and ceremonies in the Agile Scrum Framework and Explain the importance of a Sprint in Agile projects?

Roles in Agile Scrum Framework:

- **Product Owner:** Responsible for defining requirements and prioritizing the product backlog.
- **Scrum Master:** Ensures the team follows scrum practices and removes obstacles.
- **Development Team:** Designs, develops and tests the product increment during the sprint.

Ceremonies in Agile Scrum Framework:

- **Sprint Planning:** The team decides what work will be done in the upcoming sprint.
- **Daily Scrum:** A short daily meeting to discuss progress and blockers.
- **Sprint Review:** The team demonstrates the completed work to stakeholders for feedback.
- **Sprint Retrospective:** The team discusses what went well and what can be improved for the next sprint.

Importance of a Sprint in Agile projects:

- Short development cycle, usually 2 to 4 weeks.
- Delivers working product increments quickly.
- Enables continuous feedback from stakeholders.
- Helps detect and fix defects early.
- Improves team focus and productivity.
- Allows quick adaptation to changes.
- Reduces project risk.
- Ensures regular progress tracking.

5. What is a Requirement Traceability Matrix (RTM)? Prepare a sample RTM for a Registration Form with at least three requirements and their mapped test cases.

- RTM is a document that maps requirements with test cases to ensure all requirements are tested.
- Helps in impact analysis when requirements change.
- Provides clear visibility to stakeholders.
- Useful for compliance and audits in industries like banking, healthcare, aviation.

RTM for a Registration Form:

Requirement ID	Requirement Description	Test Case ID(s)	Test Case Description	Execution Status
R1	User must enter valid email address	TC1, TC2	● Enter valid email(abc@gmail.com) ● Enter invalid email(abc@)	Pass
R2	Password must be minimum 8 characters	TC3, TC4	● Enter password with 8+ characters ● Enter password with less than 8 characters	Pass
R3	All mandatory fields should not be empty	TC5, TC6	● Submit form with empty name field ● Submit form with empty email field	Fail